

Davide Adamo

PhD candidate in Applied Mathematics
Univeristè Côte d'Azur · CNRS · Inria · UMR7264 CEPAM
Based in Nice (FR)

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Profile

I am a PhD in Applied Mathematics at the University of Côte d'Azur and CNRS.

My research focuses on developing and applying machine learning methodologies to solve complex real-world problems, with a particular interest in 3D shape analysis and geometric data. With a background in mathematics and financial mathematics, my research lies at the intersection of statistical learning, geometric machine learning, and optimal transport. I develop methods for learning geometrically meaningful representations, matchings, and transformations while incorporating the underlying structure and invariances of complex data. I am particularly interested in bridging mathematical and statistical foundations with modern machine learning methodologies and translating novel research ideas into effective solutions for real-world applications.

Education

- **PhD in Applied Mathematics** *2023-26*
  3iA Côte d'Azur · CNRS · Inria · CEPAM UMR7264 Nice - Sophia Antipolis (FR)
  Thesis: *From Archaeozoology to fundamental AI: advances in Multiple Kernel Learning and Geometric Optimal Transport*
  Under the supervision of Marco Corneli, Emmanuelle Vila and Manon Vuillien

- **Master in Applied Mathematics** *2020-23*
  University of Verona Verona (IT)
  Thesis: *A Topological Data Analysis approach to classification of archaeozoological bones*

- **Bachelor in Economic & Financial Mathematics** *2017-20*
  University of Verona Verona (IT)
  Thesis: *Algoritmi per la risoluzione di FBSDEs basati sul Deep Learning*

Professional Experience

- **Research Engineer** *2023*
  CNRS · UMR7264 CEPAM Nice (FR)
  Development and testing of modern machine learning approaches for 3D challenges.
  Introduction of geometric learning approaches, such as PointNet++, in archaeozoological context.

- **Master Internship** *2022*
  Inria Sophia Antipolis (FR)
  Development of topology based descriptors for 3D scan of real-world archaeological bones.
  I became familiar with Topological Data Analysis and computational homology notions.

- **Academic Tutor** *2020-22*
  Accademia d'Arte Circense Verona (IT)
  Didactic support high/middle school for artists-students living in the academy. Daily supervision of school progress and management of school-family relationship.

Research Activity

PUBLICATIONS

- **D. Adamo**, M. Corneli, M. Vuillien and E. Vila. *Neural approximation of Procrustes-Wasserstein transport maps.* (Submitted to TMLR - Preprint 2026)
- **D. Adamo**, M. Vuillien, E. Vila and M. Corneli. *Rethinking Multiple Kernel Learning under the lenses of Importance Weighted Monte Carlo Variational Inference.* (Submitted to Journal of Classification - Preprint 2025)
- **D. Adamo**, M. Corneli, M. Vuillien and E. Vila. *An in depth look at the Procrustes–Wasserstein distance: properties and barycenters.* In International Conference on Machine Learning (ICML 2025)
- M. Vuillien*, **D. Adamo***, E. Vila, A. Agraw, T. Argant, D. Helmer, M. Mashkour, A. Moussous, O. Notter, E. Rossoni-Notter, I. Théry and M. Corneli. *Topological data analysis and multiple kernel learning for species identification of modern and archaeological small ruminants.* In Journal of Computer Applications in Archaeology (JCAA 2025)

TALKS & POSTERS

- 2026 (April - poster): *Neural approximation of Procrustes-Wasserstein transport maps*, Statlearn (Grenoble, France)
- 2025 (November): *Archaeozoology & Machine Learning: a promising matching*, Invited speaker at the 10èmes Rencontres de Statistique de l'UBS - Science des données, Histoire & Territoires (Vannes, France)
- 2025 (November - poster): *An in depth look at the Procrustes-Wasserstein distance: properties and barycenters*, SophIA Summit 2025 (Nice, France)
- 2025 (July- poster): *An in depth look at the Procrustes-Wasserstein distance: properties and barycenters*, ICML 2025 (Vancouver, Canada)
- 2025 (June): *An in depth look at the Procrustes-Wasserstein distance: properties and barycenters*, 56ièmes Journées de Statistique de la SFdS (Marseille, France)
- 2025 (May): *Optimal transport for tracking the morphological evolution of small ruminants*, Colloque des Doctorants de 2ème année de l'EDSFA (Nice, France)
- 2025 (May): *Tracking the morphological evolution of small ruminants through Wasserstein barycenters*, International Conference on Computer Application in Archaeology (Nice, France)
- 2025 (February): *Rethinking Multiple Kernel Learning under the lenses of Importance Weighted Monte Carlo Variational Inference*, Long talk at 3IA monthly seminar (Nice, France)
- 2024 (November - poster): *Rethinking Multiple Kernel Learning under the lenses of Importance Weighted Monte Carlo Variational Inference*, SophIA Summit 2024 (Nice, France)
- 2024 (May): *Rethinking Multiple Kernel Learning under the lenses of Importance Weighted Monte Carlo Variational Inference*, 55ièmes Journées de Statistique de la SFdS (Bordeaux, France)
- 2023 (November): *Astragalus bones identification via topological data analysis*, 1st International Conference on artificIAl Intelligence and applied MAthematics for History and Archaeology (IAMAHA) (Nice, France)

SUMMER SCHOOLS

- (2025) Generative Modeling Summer School (GEMSS) / Statlearn, in Nice, France
- (2024) Generative Modeling Summer School (GEMSS) in Eindhoven, Netherlands

REVIEWER ACTIVITY

- Reviewer for ICML : 2026 (**Silver Reviewer Award**)
- Reviewer for NeurIPS : 2026 (**Top Reviewer – Main Track**)

Computer Science Competences

Programming languages: Python, Matlab, R, C, C++
Frameworks: Pytorch, Tensorflow, Keras, Scikit-learn
Office: LaTeX, MS Office

Languages

Italian: Native
English: Advanced
French: Advanced
German: Basic